

Roll No:CO-2105 and Name:Sandhya Raghunath Magadum

In []:

Operators & Flow control

```
In [1]: #Operators
#1.Arithmetic operator
a = 10
b = 3

print(a + b)    # 13
print(a - b)    # 7
print(a * b)    # 30
print(a / b)    # 3.333...
print(a // b)   # 3
print(a % b)    # 1
print(a ** b)   # 1000
```

```
13
7
30
3.3333333333333335
3
1
1000
```

```
In [2]: #2.relational operators
a = 10
b = 5

print(a == b)   # False
print(a != b)   # True
print(a > b)     # True
print(a < b)     # False
print(a >= b)    # True
print(a <= b)    # False
```

```
False
True
True
False
True
False
```

```
In [3]: #2.relational operators
a = 10
b = 5

print(a == b)   # False
print(a != b)   # True
print(a > b)     # True
print(a < b)     # False
```

```
print(a >= b)    # True
print(a <= b)    # False
```

False
True
True
False
True
False

```
In [4]: #2. or
a = 10
b = 5

print(a > 5 or b > 10)    # True (one condition is true)
print(a < 5 or b > 10)    # False
```

True
False

```
In [5]: #3. not
a = 10

print(not (a > 5))        # False (because a > 5 is True)
print(not (a < 5))        # True
```

False
True

```
In [6]: #4.Bitwise operators
a = 5    # 0101 in binary
b = 3    # 0011 in binary

print("a & b =", a & b)    # AND
print("a | b =", a | b)    # OR
print("a ^ b =", a ^ b)    # XOR
print("~a =", ~a)          # NOT
print("a << 1 =", a << 1)  # Left shift
print("a >> 1 =", a >> 1)  # Right shift
```

a & b = 1
a | b = 7
a ^ b = 6
~a = -6
a << 1 = 10
a >> 1 = 2

```
In [7]: #5.assignment operator
x = 10

# =
x = 10
print("= :", x)

# +=
x += 5
print("+= :", x)
```

```

# -=
x -= 3
print("-= :", x)

# *=
x *= 2
print("*= :", x)

# /=
x /= 4
print("/= :", x)

# //=
x //= 2
print("//= :", x)

# %=
x %= 2
print("%= :", x)

# **=
x **= 3
print("**= :", x)

```

```

= : 10
+= : 15
-= : 12
*= : 24
/= : 6.0
//= : 3.0
%= : 1.0
**= : 1.0

```

In [8]: *#5.Bitwise Assignment Examples*

```

a = 5

a &= 3
print(a) # 1

a = 5
a |= 3
print(a) # 7

a = 5
a ^= 3
print(a) # 6

a = 5
a <<= 1
print(a) # 10

a = 5
a >>= 1
print(a) # 2

```

1
7
6
10
2

```
In [9]: #6.special operator
# a. identify operators
x = [1, 2, 3]
y = x
z = [1, 2, 3]

print(x is y)      # True
print(x is z)      # False
print(x is not z)  # True
```

True
False
True

```
In [10]: # b. membership operator
numbers = [10, 20, 30, 40]

print(20 in numbers)
print(50 in numbers)

print(20 not in numbers)
print(50 not in numbers)
```

True
False
False
True

Operator precedence

◆ BODMAS (used in UK, India, etc.)

B – Brackets

O – Orders (powers, square roots, exponents)

D – Division

M – Multiplication

A – Addition

S – Subtraction

```
In [11]: # operator precedence
# 1. Parentheses ()
print((10 + 5) * 2)      # 30
```

```
# 2. Exponentiation **
print(2 ** 3)          # 8

# 3. Unary Operators +, -, ~
print(-5)              # -5
print(+5)              # 5
print(~5)              # -6

# 4. Multiplication, Division, Floor Division, Modulus
print(10 * 2)           # 20
print(10 / 2)           # 5.0
print(10 // 3)          # 3
print(10 % 3)           # 1

# 5. Addition, Subtraction
print(10 + 5)           # 15
print(10 - 5)           # 5

# 6. Left Shift, Right Shift
print(5 << 1)           # 10
print(5 >> 1)           # 2

# 7. Bitwise AND
print(5 & 3)            # 1

# 8. Bitwise XOR
print(5 ^ 3)            # 6

# 9. Bitwise OR
print(5 | 3)            # 7

# 10. Relational Operators
print(10 > 5)           # True
print(10 == 10)         # True

# 11. Logical NOT
print(not True)         # False

# 12. Logical AND
print(True and False)   # False

# 13. Logical OR
print(True or False)    # True
```

```

30
8
-5
5
-6
20
5.0
3
1
15
5
10
2
1
6
7
True
True
False
False
True

```

```

In [12]: #Important Precedence Example
result = 10 + 5 * 2 > 15 and not False
print(result)

```

True

```

In [13]: #complex operator precedence
z = (1 + 2j) + (3 + 4j) * 2
print(z)

```

(7+10j)

```

In [14]: #Step-by-step:
#Step 1: 8/2*3+5+(12/4)**2-6
#Step 2: (solve division inside parentheses) 4*3+5+(3.0)**2-6
#Step 3: (solve exponent) 4*3+5+9.0-6
#Step 4: (multiplication) 12+5+9.0-6
#Step 5: (left to right addition/subtraction) 17+9.0-6
#Step 6: 20.0
result=8/2*3+5+(12/4)**2-6
print(result)

```

20.0

Flow control

Conditional - 1.if , 2.if-else, 3.if-elif-else

Iterative - 1.for ,2.while

Transfer - 1.break, 2.continue,3.pass

1 Conditional Statements

Used to make decisions based on conditions.

```
In [15]: #1. if
#Executes a block only if condition is True.

x = 10
if x > 5:
    print("x is greater than 5")
```

x is greater than 5

```
In [16]: age = 20
if age >= 18:
    print("Eligible to vote")
```

Eligible to vote

```
In [17]: #2. if-else
#Chooses between two blocks.

x = 3
if x > 5:
    print("Greater")
else:
    print("Smaller or equal")
```

Smaller or equal

```
In [18]: num = 15
if num % 2 == 0:
    print("Even Number")
else:
    print("Odd Number")
```

Odd Number

```
In [19]: #3. if-elif-else
#Used for multiple conditions.

x = 7
if x > 10:
    print("Big")
elif x > 5:
    print("Medium")
else:
    print("Small")
```

Medium

```
In [20]: marks = 78
if marks >= 90:
    print("Grade A")
elif marks >= 75:
    print("Grade B")
```

```
elif marks >= 50:  
    print("Grade C")  
else:  
    print("Fail")
```

Grade B

2 Iterative (Looping) Statements

Used to repeat a block of code.

```
In [21]: #1. for loop  
         #Used when number of iterations is known.  
  
         for i in range(5):  
             print(i)
```

0
1
2
3
4

```
In [22]: total = 0  
         for i in range(1, 6):  
             total += i  
         print("Sum =", total)
```

Sum = 15

```
In [23]: #2. while loop  
         #Runs while condition is True.  
         i = 0  
         while i < 7:  
             print(i)  
             i += 1
```

0
1
2
3
4
5
6

```
In [24]: num = 5  
         while num > 0:  
             print(num)  
             num -= 1  
         print("Blast Off!")
```

5
4
3
2
1
Blast Off!

Practice

```
In [25]: #1. if Statement
# Store a number
num = 15

# Check condition
if num > 10:
    print("Number is greater than 10")
```

Number is greater than 10

```
In [26]: #2. if-else Statement
# Store age
age = 17

# Check eligibility
if age >= 18:
    print("Eligible to vote")
else:
    print("Not eligible to vote")
```

Not eligible to vote

```
In [27]: #3. if-elif-else Statement
# Store marks
marks = 72

# Check grade
if marks >= 90:
    print("Grade A")
elif marks >= 75:
    print("Grade B")
elif marks >= 50:
    print("Grade C")
else:
    print("Fail")
```

Grade C

```
In [28]: #4. for Loop
# Print numbers from 1 to 5

for i in range(1, 9):
    print(i)
```

1
2
3
4
5
6
7
8

```
In [29]: #5. while Loop
# Initialize variable
i = 1

# Run loop while condition is true
while i <= 5:
    print(i)
    i += 1
```

1
2
3
4
5

In []: